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# The responses of photosynthetic rate of dominant species to extreme drought in Inner Mongolia grasslands

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## Abstract

Estimating the effects of extreme drought on the photosynthetic rates (Pn) of dominant plant species is crucial for understanding the mechanisms driving the impacts of extreme drought on ecosystem functioning. Extreme drought may result from either reduced rainfall amount or decreased rainfall frequency, and the impacts of different patterns of extreme drought may vary greatly. In addition, different grasslands likely appear various sensitivity to different extreme drought patterns. However, there have been no reports on the effects of different extreme drought patterns on dominant species Pn in different grassland types. Here, we conducted multi-year extreme drought simulation experiments (reducing each rainfall event by 66% during the growing season, CHR *vs*. completely excluding rainfall during a shorter portion of the growing season, INT) in two different grasslands (desert grassland *vs*. typical grassland) from 2014. The Pn of two dominant species in each grassland was measured in July and August 2017. Both CHR and INT significantly decreased dominant species Pn, with INT causing more negative impacts on Pn regardless of grassland types. The response ratios of Pn in desert grassland were generally higher than that of typical grassland, especially for *Leymus chinensis* in CHR. These results demonstrated that decreased rainfall frequency had a more negative effect on Pn compared with reduced rainfall amount, with grassland types changing the magnitude, but not the direction, of the effects of extreme drought patterns. These findings highlight the importance of considering extreme drought patterns and grassland types in ecosystem management in the face of future extreme droughts.

***Keywords:*** climate extremes, desert grassland, typical grassland, extreme drought pattern, seasonal drought

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内蒙古草原优势物种光合速率对极端干旱的响应

摘要：评估极端干旱对优势植物物种光合速率(Pn)的影响对于我们理解极端干旱影响生态系统功能的机制至关重要。极端干旱可能由降雨量减少或降雨频率减少造成。这两种不同模式的极端干旱可能造成不同的影响。此外，不同草原类型可能对这两种极端干旱模式表现出不同的敏感性。然而，目前尚不清楚这两种极端干旱模式对草原优势植物物种Pn的影响有何差异，以及在不同草原类型这两种不同模式的极端干旱的效应是否改变。为此，我们于2014年在内蒙古草原两个不同类型的草原(荒漠草原*vs*.典型草原)站点各进行了一项多年的极端干旱(CHR：将生长季期间每次降雨事件的降雨量减少66% *vs*. INT：完全排除生长季内一段较短时间的降水)模拟实验。于2017年的7月和8月，我们在这两个草原站点各测量了两种优势物种的Pn。结果发现，CHR和INT均显著降低了优势物种的Pn。无论草原类型如何，INT对Pn造成的负面影响都更大。在荒漠草原，极端干旱对Pn的负面影响普遍大于典型草原，尤其是对于CHR处理下的羊草(*Leymus chinensis*)。这些结果表明，与降雨量减少相比，降雨频率减少所引起的极端干旱对Pn的负面影响更大。草地类型可能会改变极端干旱模式效应的程度，但不会改变其方向。这些发现强调了应对未来极端干旱的草原生态系统管理应考虑极端干旱模式和草地类型的影响。

关键词：极端气候，荒漠草原，典型草原，极端干旱模式，季节性干旱

## INTRODUCTION

Extreme droughts are defined as precipitation events that fall below the 5th percentile of the historical range of variability (Smith 2011). They have the potential to cause widespread importance on ecosystem functions and processes (Hopkins and Del Prado 2007; Ma *et al.* 2023; Smith 2011) through affecting plants growth and production (Dumont *et al.* 2015; Knapp *et al.* 2015; Wei *et al.* 2022). Extreme drought may result from either reduction in rainfall amount or decrease in rainfall frequency, and the impacts of different patterns of extreme drought on ecosystems may vary greatly (Goke and Martin 2023). Clarifying the differences in the impacts of different extreme drought patterns is important for ecosystem management in the face of future extreme droughts.

The photosynthesis is the basis for plant growth and production (Garcia *et al.* 2023). The different extreme drought patterns might cause different negative effects on plant photosynthetic rates (Pn) (Li *et al.* 2024). However, most studies have primarily focused on the drought caused by reduced rainfall amount, while the effects of drought resulted from a decrease in rainfall frequency have been relatively neglected (Luo *et al.* 2023; Sun *et al.* 2023). With the progression of climate change, both the frequency and magnitude of extreme drought have markedly increased (Ma *et al.* 2023; Wei *et al.* 2022). Extreme drought events may persist for several weeks or even months, leading to a pronounced alteration in frequency (Ma *et al.* 2023), underscoring the significance of studying rainfall frequency. Currently, research on the ecological responses to extreme drought patterns has predominantly centered on biomass or primary productivity (Carroll *et al.* 2021), with less emphasis on comparing the effects of extreme drought characterized by a decrease in rainfall frequency or reduction in rainfall amount on Pn.

Grassland ecosystems inherently exhibit complexity due to the presence of diverse biotic components such as species assemblages, and abiotic components including soil, precipitation and temperature, along with their associated processes (Jaman *et al.* 2022; Lüscher *et al.* 2022; Stuart-Haëntjens *et al.* 2018). Consequently, the impacts of extreme drought can vary significantly across distinct grassland ecosystems. The previous studies suggest two distinct hypotheses regarding the response of different ecosystems to extreme drought (Hoover *et al.* 2021; Ilyas *et al.* 2021). One hypothesis highlighted that wetter locations are more sensitive than drier ones, as plants in arid regions are accustomed to enduring water stress and have adapted to drought conditions (Byrne *et al.* 2013; Grime 1997; Tielbörger *et al.* 2014). The other claimed that drier locations are more sensitive than wetter ones, since in moist environments, water is not the primary limiting factor as it is in arid ecosystems (Huxman *et al.* 2004; Knapp *et al.* 2015; Stuart-Haëntjens *et al.* 2018). The debates stimulated comparative studies of the effects of extreme drought in different grasslands in order to find general rules (Knapp *et al.* 2015; Yahdjian *et al.* 2021). The effects of extreme drought on photosynthetic rate may be a potential mechanism

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underlying the differential responses of different grasslands to extreme drought (Jia *et al.* 2023; Knapp *et al.* 2020; Yang *et al.* 2023). However, knowledge about how photosynthetic rates respond to extreme drought patterns in different grassland ecosystems remains elusive.

The mechanisms driving drought effects on Pn have been extensively investigated (Li *et al.* 2023; Vitale *et al.* 2009). Drought can lead to the closure of leaf stomata and a decrease in transpiration (Li *et al.* 2023; Vitale *et al.* 2009), disrupting the water balance within plants (Flexas *et al.* 2006) and then affecting the Pn. Stomatal closure can reduce stomatal conductance and vapor pressure deficit, potentially hindering the assimilation of carbon dioxide (Tor-ngern *et al.* 2021). Additionally, the carbon dioxide produced during plant dissimilation may not be efficiently released, further impairing the photosynthetic process (Ort and Baker 2002). The impact of extreme drought on photosynthesis may exhibit nonlinear behavior due to physiological trade-offs, introducing uncertain effects on the photosynthetic process (Flexas *et al.* 2006; Vitale *et al.* 2009). Therefore, establishing the relationship between Pn and photosynthetic traits can provide a comprehensive assessment of Pn response to extreme drought.

The Inner Mongolia grassland is experiencing an escalation in both the frequency and intensity of extreme drought events (Meehl *et al.* 2007) which threatens local ecosystem functions and services. Thus, there is an urgent need to investigate the effects of extreme drought on these grassland ecosystems. In this study, we assess the responses of the photosynthesis rates (Pn) of dominant plant species to chronic drought (CHR, reducing each rainfall event by 66% during the growing season) and intense drought (INT, completely excluding rainfall during a shorter portion of the growing season) in two distinct grasslands (desert grassland *vs*. typical grassland) in the Inner Mongolia grassland. We addressed the following three questions: (i) how extreme droughts affect dominant species Pn; (ii) whether extreme drought patterns influence the responses of Pn; (iii) whether Pn appeared varied sensitivity to extreme drought patterns in desert grassland and typical grassland.

## MATERIALS AND METHODS

### Experimental sites

This study was conducted at two distinct sites: one located in *Sheila MuRen* (111°53′ E, 41°47′ N), characterized by a desert grassland environment, and the other in *Xilingol* (116°42′ E, 43°38′ N), representing a typical grassland habitat. *Sheila MuRen* experiences a mean annual precipitation of 236 mm, with a mean temperature of 3.4°C and an annual potential evaporation rate of 2227 mm. The soil is classified as chestnut soil, containing 2%–4% soil organic matter, 0.1%–0.15% soil total nitrogen, 0.08%–0.15% soil total phosphorus and 2%–2.5% soil total potassium, with a pH range of 8–8.5. Dominant species in this area include *Stipa krylovii*, *Caragana stenophylla*, *Leymus chinensis*, *Artemisia frigida*, *Caragana microphylla* and *Cleistogenes squarrosa*. In contrast, *Xilingol* receives a mean annual precipitation of 346 mm, with a mean temperature of 2.3°C and an annual potential evaporation rate of 1862 mm. The soil type is also chestnut soil, containing 1.6%–3.7% soil organic matter, 0.12%–0.18% soil total nitrogen, 0.1%–0.17% soil total phosphorus and 1.8%–2.6% soil total potassium, with a pH range of 7.8–8.2. Dominant species in *Xilingol* include *L. chinensis*, *Stipa grandis*, *Achnatherum sibiricum*, *Agropyron cristatum*, *C. microphylla* and *Salsola collina*.

### Experimental design

The experiments were set up in 2014, and the treatments have been conducted since 2015. There were three treatments in each site, i.e. control (CON), chronic drought (CHR) and intense drought (INT) with six replicates. CHR reduced each rainfall event by 66% during the growing season (from May to August), while INT completely excluded rainfall from 1 June to 31 July (Luo *et al.* 2022). The size of the shelters is 6.2 m × 6.2 m, which is fully covered the size of the experimental plots (6 m × 6 m) (Griffin-Nolan *et al.* 2019; Yahdjian and Sala 2002). The roof (consist of clear acrylic with the characteristic of high light transmission, low yellowness index and UV transparent) of each modular unit will be slanted at 15° toward the edge of the plot, and the lower edge will be oriented to the prevailing wind direction to minimize blow-in (Griffin-Nolan *et al.* 2019). In order to reduce the marginal effect, we only use the middle area (4 m × 4 m) as the sampling area. To prevent water exchange between individual plots, we implemented hydrological isolation by installing waterproof plastic boards along the boundaries of each plot to a depth of 1 m.

### Data measurements

We measured monthly precipitation and Pn in 2017. Precipitation was recorded by the weather station near

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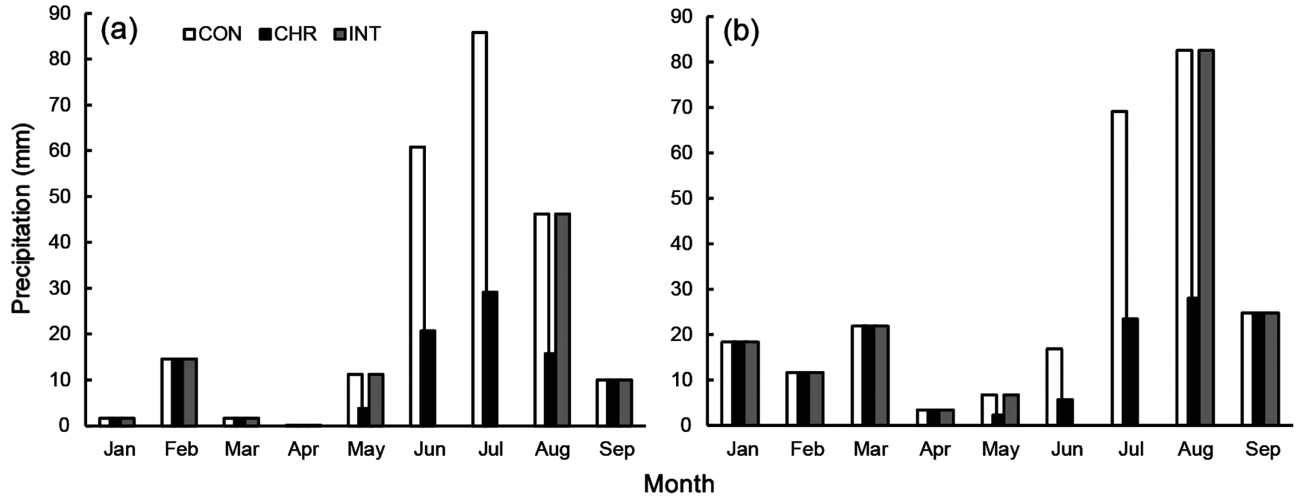



**Figure 1:** The monthly precipitation in *Sheila MuRen* (a) and *Xilingol* (b) in 2017. Abbreviations: CON = control, CHR = chronic drought (reducing each rainfall event by 66% from May to August), INT = intense drought (completely excluding rainfall from 1 June to 31 July).

each experiment site. In *Sheila MuRen*, the precipitation for the CHR and INT treatments was 97.26 and 85.30 mm, respectively (Fig. 1a). The reductions in precipitation for the CHR and INT treatments were 134.64 and 146.60 mm, respectively. In *Xilingol*, the precipitation for the CHR and INT treatments were 139.90 and 169.60 mm, respectively (Fig. 1b). The reductions in precipitation of CHR and INT were 115.70 and 86.00 mm, respectively. We measured Pn at the beginning and end of July and August, with the mean value of twice measurements each month representing the level of Pn in that month. The two most dominant species were selected in each site, i.e. *S. krylovii* and *L. chinensis* in *Sheila MuRen* and *S. grandis* and *L. chinensis* in *Xilingol*. The selected species at different sites belonged to the same species or genus to minimize the impact of species-specific responses on the assessment of extreme drought effects. For each species, four healthy and representative plant individuals were selected randomly in each plot, and each plant was measured three times with three different leaves between 11:00 and 13:00, using a Portable Photosynthesis System (Li-6400, Lincoln, NE, USA) with red and blue artificial light source. Measuring during the midday period allowed us to focus on maximum photosynthetic activity and avoid potential confounding factors associated with morning or afternoon extremes. All indices were measured during periods characterized by the absence of precipitation events on the preceding day and clear sky conditions on the day of measurement. The target plants were labeled with flags to ensure that the same plants were measured each time. The measurements were performed with the photosynthetic photon flux density of 1500 μmol $m^{-2}$ $s^{-1}$, the $CO_2$ concentration of 400 μmol $mol^{-1}$ and the leaf temperature of 30°C. The photosynthetic photon flux density closely approximated the light intensity experienced by plants in full sunlight on a clear day which ensured that photosynthesis is operating at its maximum rate. At a leaf temperature of 30°C, most plant species exhibited optimal photosynthetic activity, which ensured that photosynthesis is not limited by temperature. The infrared gas analyzer simultaneously recorded the following four photosynthetic traits: transpiration rate (Tr, mmol $H_2O$ $m^{-2}$ $s^{-1}$), vapor pressure deficit ($V_{pdl}$, Pa), intercellular $CO_2$ concentration ($C_i$, μmol $CO_2$ $mol^{-1}$) and stomatal conductance ($G_s$, mol $H_2O$ $m^{-2}$ $s^{-1}$).

## Data analysis

The response ratios (RR) of Pn to extreme drought was defined as $RR = Pn_{(CON\text{-}Treatments)}/Pn_{(CON)}$, treatments represent CHR or INT. The sensitivity of Pn represents the change of photosynthetic rate caused by per unit of precipitation ($P$) variation, i.e. $Sensitivity = Pn_{(CON\text{-}Treatments)}/P_{(CON\text{-}Treatments)}$. The formula $WUE = Pn/Tr$ is used to calculate the water use efficiency. Repeated measures analysis of variance were used to analyze the difference of Pn, the RR and the sensitivity among treatments, between months and sites. Significant differences were estimated at the level of 1% ($P < 0.01$) using Tukey's test. Linear regression equations or quadratic equations were used to determine the relationship between Pn and photosynthetic traits (Tr, $V_{pdl}$, $C_i$, $G_s$ and WUE). All the statistical analyses were conducted using SPSS software (SPSS v.19; IBM, Chicago IL, USA).

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## RESULTS

### Effects of extreme drought on plant photosynthetic rates

Both chronic drought (CHR) and intense drought (INT) significantly decreased photosynthetic rates (Pn) across the species and months in both the desert and typical grasslands (Fig. 2). The Pn under INT were generally lower than those under CHR (Fig. 2). On average, Pn was decreased by 35.43% and 45.21% in CHR and INT, respectively. The Pn in August was generally lower than those in July (all $P < 0.01$).

The RR of Pn in INT consistently exceeded that in CHR (Fig. 3). The RR of Pn in *Sheila MuRen* was higher than that in *Xilingol* ($P < 0.01$). On average, the RR of Pn in *Sheila MuRen* and *Xilingol* was 0.50 and 0.36, respectively. The RR in August was generally higher than that in July (all $P < 0.01$, except for *L. chinensis* in *Sheila MuRen*). The differences in Pn sensitivity between sites and months exhibited opposing patterns compared with the differences in RR (Fig. 4), while the extreme drought patterns had similar effects on Pn and RR across species.

### The effect of extreme drought on the relationships between photosynthetic rates and photosynthesis traits

In each treatment of two sites, linear correlations occurred between the Pn and transpiration rate (Tr),

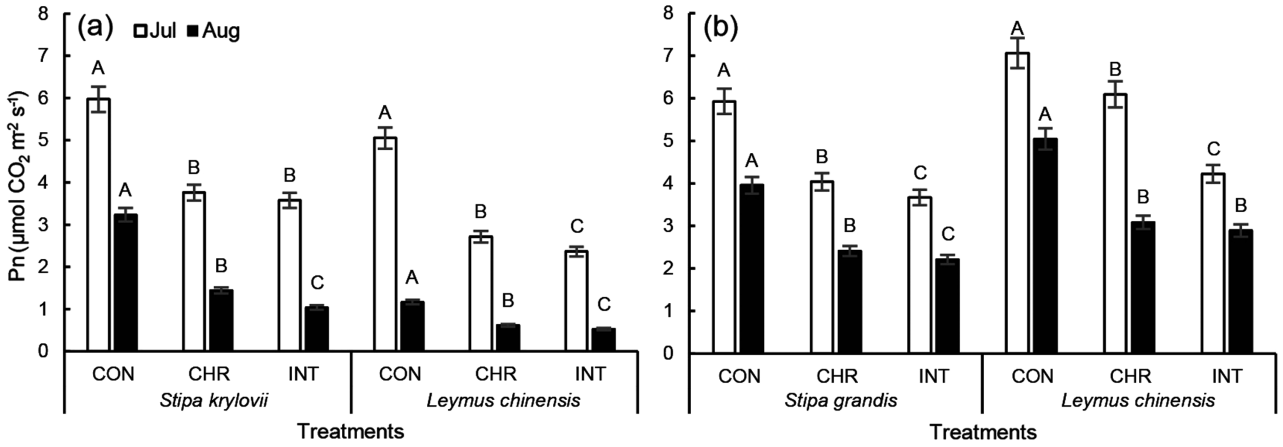



**Figure 2:** The effects of extreme drought on photosynthetic rates (Pn) in *Sheila MuRen* (a) and *Xilingol* (b). Different capital letters indicate significant differences among treatments ($P < 0.01$). Abbreviations: CON = control, CHR = chronic drought (reducing each rainfall event by 66% from May to August), INT = intense drought (completely excluding rainfall from 1 June to 31 July). Error bars represent mean ± SE.

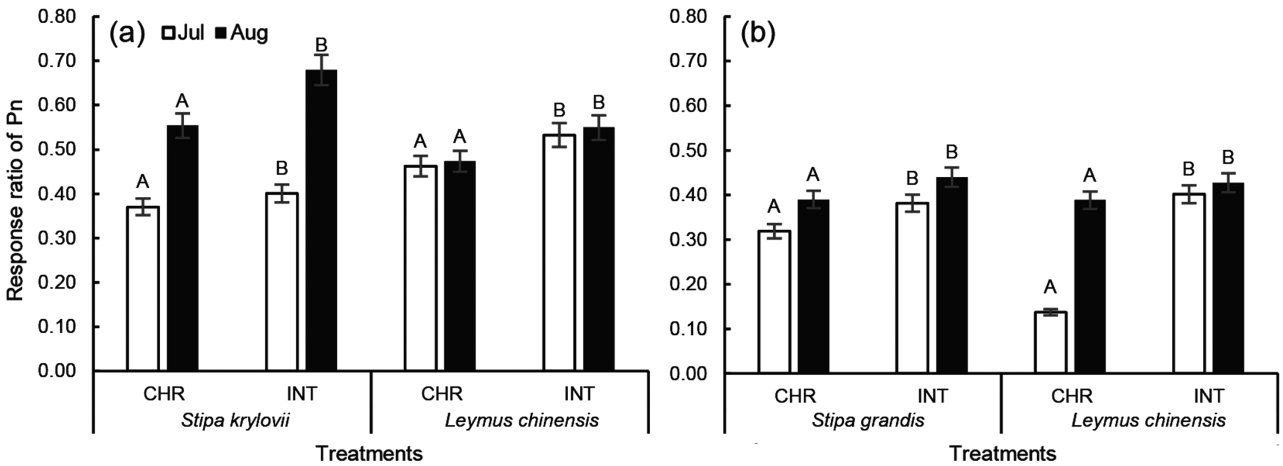



**Figure 3:** The response ratios (RR) of photosynthetic rates (Pn) to extreme drought in *Sheila MuRen* (a) and *Xilingol* (b). Different capital letters indicate significant differences among treatments ($P < 0.01$). Abbreviations: CHR = chronic drought (reducing each rainfall event by 66% from May to August), INT = intense drought (completely excluding rainfall from 1 June to 31 July). Error bars represent mean ± SE.

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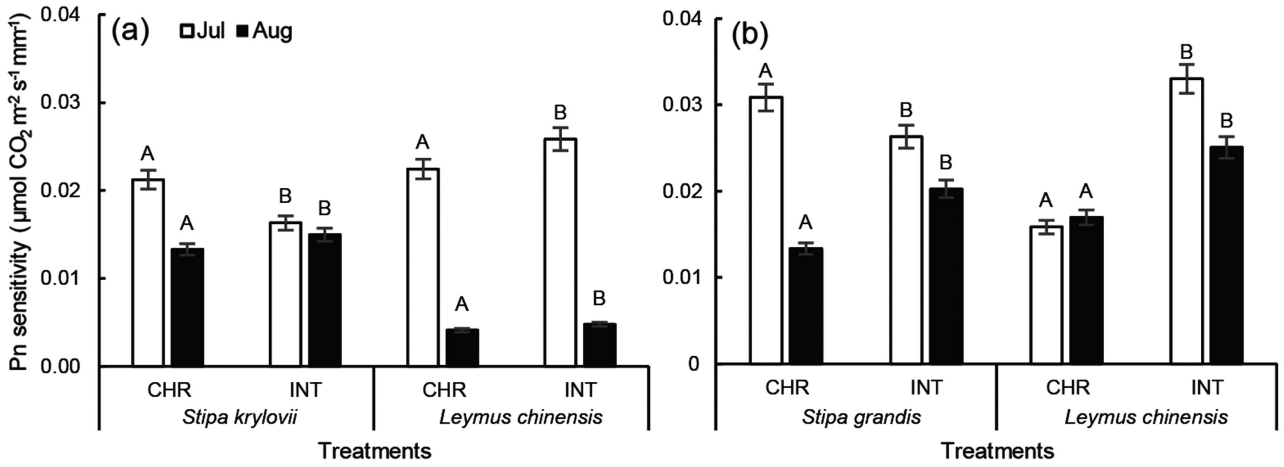



**Figure 4:** Effect of extreme drought on the sensitivity of photosynthetic rates (Pn) in *Sheila MuRen* (a) and *Xilingol* (b). Different capital letters indicate significant differences among treatments ($P < 0.01$). Abbreviations: CHR = chronic drought (reducing each rainfall event by 66% from May to August), INT = intense drought (completely excluding rainfall from 1 June to 31 July). Error bars represent mean ± SE.

between Pn and vapor pressure deficit ($V_{pdl}$, except the INT-*Leymus* in *Sheila MuRen*), between Pn and intercellular $CO_2$ concentration ($C_i$, except the CHR-*Stipa* in *Sheila MuRen* and *Xilingol*), between the Pn and stomatal conductance ($G_s$), between Pn and water use efficiency (WUE, except the CHR-*Stipa* in *Sheila MuRen*, the CHR-*Stipa*, CHR-*Leymus* and INT-*Leymus* in *Xilingol*) (Table 1). In different treatments, the slopes of the linear equation were different and mostly showed the pattern: INT > CHR > CON (Table 1).

## DISCUSSION

There is increasing evidence that the frequency and severity of extreme drought have already increased in many regions (Spinoni *et al.* 2018). It is not just the decrease in rainfall amount, the frequency of rainfall events has also seen a severe reduction. In the grasslands of Inner Mongolia, both the number and duration of extreme drought events have increased significantly, with even previously humid areas transitioning to dry conditions (An 2016; Di *et al.* 2022). While previous studies have primarily focused on the effects of extreme drought resulted from a reduction in the amount of rainfall event, there has been limited research on the impacts of decreased rainfall frequency (Huang *et al.* 2023; Zhang *et al.* 2016). In this study, we conducted two extreme drought treatments (reducing each rainfall event by 66% from May to August, CHR *vs.* completely excluding rainfall from 1 June to 31 July, INT) and measured the responses of plant photosynthetic rates (Pn). We found that INT consistently had a greater impact on Pn compared with CHR, despite the INT treatment experiencing a weaker reduction in rainfall (Fig. 1). This suggests that a decrease in rainfall frequency may improve extreme drought effects on Pn. The previous study showed that INT caused more negative impacts on plant community productivity compared with CHR in Konza Prairie Biological Station (Carroll *et al.* 2021). Our study providing further clues to plant physiological responses to explain the effects of extreme drought patterns. Overall, these results highlight the importance of considering extreme drought patterns in ecosystem management and conservation efforts.

Previous studies have indicated that the effects of extreme drought on Pn vary across ecosystems and are often nonlinear. For instance, Gao *et al.* (2019) observed that the Pn of *Festuca arundinacea* remained unchanged when precipitation decreased by 50% but decreased significantly by 51.9% when precipitation decreased by 70% in a relatively mesic ecosystem. In contrast, another study showed a different pattern in a relatively arid ecosystem, where the Pn of *Lolium multiflorum* decreased by 30.7% with a 50% reduction in precipitation, and by 45.0% with a 70% reduction in precipitation (Li *et al.* 2018). Thus, while low-intensity drought may elicit varying Pn responses in mesic and arid ecosystems, extreme drought is likely to induce similar responses in Pn across different ecosystems. This finding corresponded well with previous studies that deserts, grasslands and forests exhibit the same rate of biomass reduction when water is extremely

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**Table 1:** The effects of extreme drought on the relationships between photosynthetic rates (Pn) and photosynthesis traits

| Sites | Physiological factors | Tr ($N = 91$) | $V_{pdl}$ ($N = 90$) | $C_i$ ($N = 85$) | $G_s$ ($N = 67$) | WUE ($N = 80$) |
|---|---|---|---|---|---|---|
| *Sheila MuRen* | CON-*Stipa* | $y = 2.64x + 1.41$<br>$R^2 = 0.70$, $P = 0.00$ | $y = 1.48x + 2.21$<br>$R^2 = 0.25$, $P = 0.00$ | $y = -0.05x + 16.77$<br>$R^2 = 0.20$, $P = 0.00$ | $y = 70.75x + 3.09$<br>$R^2 = 0.38$, $P = 0.00$ | $y = 1.58x + 2.94$<br>$R^2 = 0.29$, $P = 0.00$ |
| | CHR-*Stipa* | $y = 2.84x + 4.44$<br>$R^2 = 0.47$, $P = 0.00$ | $y = -1.17x + 14.00$<br>$R^2 = 0.27$, $P = 0.00$ | $y = -0.0001x^2 + 0.04x + 7.31$<br>$R^2 = 0.26$, $P = 0.01$ | $y = 108.44x + 0.59$<br>$R^2 = 0.50$, $P = 0.00$ | $y = -0.13x^2 + 2.17x + 1.16$<br>$R^2 = 0.24$, $P = 0.15$ |
| | INT-*Stipa* | $y = 3.59x + 2.72$<br>$R^2 = 0.26$, $P = 0.00$ | $y = -2.56x + 17.02$<br>$R^2 = 0.35$, $P = 0.00$ | $y = -0.007x + 7.85$<br>$R^2 = 0.11$, $P = 0.001$ | $y = 115.27x + 1.03$<br>$R^2 = 0.65$, $P = 0.00$ | - |
| | CON-*Leymus* | $y = 0.28x + 0.37$<br>$R^2 = 0.54$, $P = 0.00$ | $y = 0.27x + 5.38$<br>$R^2 = 0.15$, $P = 0.00$ | $y = -0.008x + 8.20$<br>$R^2 = 0.27$, $P = 0.00$ | $y = 60.67x - 0.04$<br>$R^2 = 0.52$, $P = 0.00$ | $y = 1.26x + 2.71$<br>$R^2 = 0.25$, $P = 0.00$ |
| | CHR-*Leymus* | - | $y = 1.47x - 0.19$<br>$R^2 = 0.40$, $P = 0.00$ | $y = -0.13x + 7.06$<br>$R^2 = 0.42$, $P = 0.00$ | $y = 83.83x + 0.74$<br>$R^2 = 0.33$, $P = 0.00$ | $y = 0.76x + 2.41$<br>$R^2 = 0.33$, $P = 0.00$ |
| | INT-*Leymus* | $y = 0.89x + 3.07$<br>$R^2 = 0.19$, $P = 0.001$ | $y = -2.61x^2 + 20.26x - 34.07$<br>$R^2 = 0.57$, $P = 0.16$ | $y = -0.009x + 5.98$<br>$R^2 = 0.51$, $P = 0.00$ | $y = 90.71x + 0.48$<br>$R^2 = 0.32$, $P = 0.00$ | $y = 1.07x + 1.63$<br>$R^2 = 0.77$, $P = 0.00$ |
| *Xilingol* | CON-*Stipa* | $y = 2.34x + 0.43$<br>$R^2 = 0.85$, $P = 0.00$ | $y = 1.36x - 2.28$<br>$R^2 = 0.33$, $P = 0.00$ | $y = -0.01x + 5.81$<br>$R^2 = 0.21$, $P = 0.00$ | $y = 68.85x + 3.04$<br>$R^2 = 0.25$, $P = 0.01$ | $y = 1.12x + 1.78$<br>$R^2 = 0.20$, $P = 0.00$ |
| | CHR-*Stipa* | $y = 2.58x + 1.49$<br>$R^2 = 0.68$, $P = 0.00$ | - | $y = -0.0002x^2 + 0.07x + 0.28$<br>$R^2 = 0.21$, $P = 0.02$ | $y = 70.75x + 3.10$<br>$R^2 = 0.38$, $P = 0.00$ | $y = -0.89x^2 + 6.98 - 4.92$<br>$R^2 = 0.25$, $P = 0.13$ |
| | INT-*Stipa* | $y = 3.15x - 0.40$<br>$R^2 = 0.92$, $P = 0.00$ | $y = 3.14x - 0.97$<br>$R^2 = 0.40$, $P = 0.00$ | $y = -0.02x + 7.45$<br>$R^2 = 0.33$, $P = 0.00$ | $y = 104.79x + 0.72$<br>$R^2 = 0.47$, $P = 0.00$ | $y = 1.49x + 0.57$<br>$R^2 = 0.26$, $P = 0.00$ |
| | CON-*Leymus* | $y = 2.25x + 1.00$<br>$R^2 = 0.69$, $P = 0.00$ | $y = 0.73x + 4.66$<br>$R^2 = 0.22$, $P = 0.00$ | $y = -0.016x + 7.34$<br>$R^2 = 0.26$, $P = 0.01$ | $y = 72.22x - 0.30$<br>$R^2 = 0.57$, $P = 0.00$ | $y = 1.31x + 1.59$<br>$R^2 = 0.26$, $P = 0.00$ |
| | CHR-*Leymus* | $y = 1.94x + 2.28$<br>$R^2 = 0.54$, $P = 0.00$ | $y = 1.70x - 1.00$<br>$R^2 = 0.37$, $P = 0.00$ | - | $y = 86.76x + 2.22$<br>$R^2 = 0.49$, $P = 0.00$ | $y = -0.74x^2 + 5.92x - 4.23$<br>$R^2 = 0.31$, $P = 0.09$ |
| | INT-*Leymus* | $y = 3.68x + 1.66$<br>$R^2 = 0.57$, $P = 0.00$ | $y = 1.51x + 1.23$<br>$R^2 = 0.32$, $P = 0.00$ | $y = -0.015x + 7.30$<br>$R^2 = 0.51$, $P = 0.00$ | $y = 110.97x + 0.13$<br>$R^2 = 0.46$, $P = 0.00$ | $y = -0.29x^2 + 3.80x - 3.52$<br>$R^2 = 0.49$, $P = 0.04$ |

Note: '-' represents no significant regression relationships. Abbreviations: $C_i$ = intercellular $CO_2$ concentration, $G_s$ = stomatal conductance, Tr = transpiration rate, $V_{pdl}$ = vapor pressure deficit.



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limiting (Huxman *et al.* 2004; Knapp *et al.* 2015; Wang *et al.* 2022). Our results provide physiologic information for above-ground net primary productivity (ANPP) evaluation and thus are very helpful to explore the mechanisms underlying the extreme drought impacts on ecosystem function and processes.

The impacts of CHR on Pn accumulated from July to August (Fig. 3). This could be caused by the accumulated negative effects of extreme drought on soil moisture (Hoover *et al.* 2021). This aligns with the results of a potted experiment, where after 6 days of drought resulting in a 50% reduction in soil water content, there was no change in the Pn of *Coronilla varia* (Yang *et al.* 2013). However, by the 12th day, Pn had decreased by 24.1%, and by the 18th day, it had declined by 42.5% (Yang *et al.* 2013). For the INT treatment, the decline in Pn continued to exacerbate in August, indicating the delayed effects of extreme droughts. It implies that even as climatic conditions return to normal, there likely remains a prolonged water deficit (Hoover *et al.* 2021), leading to a continual decline in photosynthetic rates following the end of the drought period.

The impact of extreme drought on Pn varied significantly among different plant species and across various sites (Flexas *et al.* 2022). Our findings revealed that, compared with the CON, extreme drought led to a substantial decrease in Pn levels: *S. krylovii* by 46.67% and *L. chinensis* by 50.01% in *Sheila MuRen* and *S. grandis* by 37.62% and *L. chinensis* by 32.72% in *Xilingol*. These diverse responses observed among species can be attributed to complex physiological and ecological processes, including trade-offs between WUE and growth rate, specific plant genotypes and long-term environmental adaptation (Smith *et al.* 2009; Xu *et al.* 2021). Furthermore, the effects of extreme drought on Pn varied between the two study sites. Notably, the impact on *L. chinensis* in *Sheila MuRen* was more pronounced compared with that in *Xilingol*. While mean annual precipitation was considered the primary factor influencing the differential responses among ecosystems, the characteristics of dominant species and plant functional diversity also played significant roles (Luo *et al.* 2021; Miao *et al.* 2022; Wang *et al.* 2023). Hence, conducting experiments across multiple sites is essential to comprehensively assess both biotic and abiotic determinants of ecosystem responses to extreme drought.

There are two primary methods for investigating the relative response of ecosystems to climate change or human disturbance. One method is sensitivity, which measures the change caused by each unit of precipitation, while the other is the RR, which compares the relative change with a control (Huxman *et al.* 2004; Knapp *et al.* 2015). Our results indicated that the sensitivity of Pn in *Sheila MuRen* (mean annual precipitation 236 mm) was generally lower than that in *Xilingol* (mean annual precipitation 346 mm), suggesting that Pn sensitivity increases with precipitation. In contrast, studies conducted in North American grasslands showed that the sensitivity of ANPP decreased with increasing precipitation (Huxman *et al.* 2004). This difference may be attributed to the different strategies employed by plants to cope with drought stress. In Inner Mongolia, ecosystems are much drier than most grasslands in North America, and plants are more constrained by trade-offs with traits for stress tolerance (Liu and Li 2020; Smith *et al.* 2009). However, an opposite pattern was found for the RR of Pn, suggesting that in arid sites, Pn was more severely affected by extreme droughts compared with relatively wet sites when compared with a control. Thus, RR and sensitivity of Pn showed completely opposite patterns in response to extreme drought in our study sites, indicating that the sensitivity of ecosystem functions may differ (even show opposite trends) depending on the method used.

Plants typically employ protective mechanisms to alleviate damage caused by extreme drought stress, allowing them to withstand harsh conditions (Ilyas *et al.* 2021; Wahab *et al.* 2022). Adjustments in transpiration rate, vapor pressure deficit, intercellular $CO_2$ concentration and stomatal conductance represent specific responses to extreme drought, providing effective strategies for combating stress (Ge *et al.* 2016; Grossiord *et al.* 2020). The relationships between a plant's Pn and its photosynthetic traits, both of which are influenced by water availability, may exhibit some variability (Gonzalez-Paleo and Ravetta 2018; Mueller *et al.* 2022). In our study, we observed that the two extreme drought patterns generally did not alter the fundamental relationship between photosynthetic traits and Pn. However, they did affect the slopes of the linear equations describing this relationship. Specifically, the slopes of the linear functions for the INT and CHR treatments were greater than that of the CON treatment, indicating that extreme drought intensified the response of Pn to photosynthetic trait factors. A closer relationship between photosynthetic traits and Pn may represent

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a double-edged sword: while it may facilitate faster recovery of plant Pn once extreme drought is alleviated, prolonged exposure to extreme drought conditions may lead to greater damage or even collapse of Pn.

## CONCLUSIONS

Aim to clarify the impact of extreme drought patterns on the photosynthetic rates (Pn) of dominant plant species in different grassland environments, we conducted experiments involving two extreme drought patterns (chronic drought: CHR, reducing rainfall by 66% from May to August; intense drought: INT, reducing rainfall by 100% from June to July) in both desert grassland and typical grassland environments. We found that both CHR and INT significantly decreased the Pn of dominant species in July and August in both types of grasslands, with Pn decreasing by an average of 35.43% and 45.21% in CHR and INT, respectively. Moreover, we observed that the RR of Pn in desert grassland was generally higher than that of typical grassland, indicating a greater sensitivity to extreme drought in desert grasslands. These findings underscore the importance of considering different drought patterns and grassland types in ecosystem management and conservation efforts. By understanding the mechanisms driving the response of dominant plant species to extreme drought, we can better inform strategies to mitigate the impacts of drought on ecosystem functions and processes in arid and semi-arid grasslands.


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